Tactix: Countercheck of the Winner PB Proof Gap

Report to Andrew

25 September 2026

Outcome

The exact Tseitin instance for which AM408 recovered a LymphoSAT PB proof is already certified by current Tactix. A pinned Release Tactix build returned UNSAT through Lucky's startup parity recognizer, emitted PBP, and passed both pinned VeriPB and CakePB against the original CNF. Thus that winner witness does not show a missing semantic refutation for this formula. The real architecture gap is narrower: Tactix has no general live PB derivation action or proof-event interface after the startup boundary.

Evidence

Hwulab job `1721399` completed `0:0` with two CPUs and 8 GiB. The exact input SHA-256 matches AM408; source was `6c176683`, and the pinned binary SHA-256 begins `2f1d91b6`. Solver exit was 20. It reported one recognized parity refutation and emitted a 116,111-byte PBP (SHA-256 prefix `df5e3736`); VeriPB reported `s VERIFIED UNSATISFIABLE` and CakePB `s VERIFIED UNSAT` on the 162,748-byte elaboration. LymphoSAT's exact baseline proof was 80,981 bytes. The different allocation and sub-centisecond Tactix timing prevent a valid speed comparison. The runner, full proof, checks, resource log and hashes are frozen in the AM410 probe under `benchmarks/probes/`; the AM410 technical report gives the exact commands and interpretation.

Decision and remaining work

Do not implement a family-specific Tseitin certificate or change canonical DSR on this evidence. A generally selectable PB surface may still be needed for winner techniques outside complete startup refutations; that requires an explicit format and proof-state contract. H2's frozen no-option DSR failure and the post-finalization C-API wrong-SAT model remain open. AM400/401 full Debug and Opus fuzz jobs are still active. N1–N4 and the retained Rank/B+C/WV gates are unaccepted, so the requested full Tier-D run at 5,000 seconds, four CPUs and 32 GiB has not been submitted. Online learning remains prototype-only pending a default-exact slot controller and owned safe-point sub-strategy switching.